Revised Proposal

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Team Pangolin: Gloria Moon, Neha Choudhari, Gus Saalfeld
Client: Dietrich College Dean's Office of Diversity, Equity, and Inclusion
Client Contact: Ayana Ledford (Associate Dean for DEI)

Community Partner Background

About the Organization

The mission of the Dietrich College Dean's Office of Diversity, Equity, and Inclusion (DEI) is to "*provide oversight on diversity, equity and inclusion issues and facilitates communication and accountability*" [1]. The office is organized into an advisory committee and an implementation committee [2], with both parties working together to bring inclusion to community programs, events, and resources. They monitor DEI initiatives within the larger Dietrich ecosystem, which is the second-largest academic body at CMU. Consequently, they reach a large audience that spans across undergraduate students, graduate students, faculty, and staff [3]. The DEI system is decentralized and dynamic. DEI relies on shared responsibility, as it attaches itself to a wide array of initiatives. They communicate with part-time volunteers who collect data on initiatives coordinated with different faculties, programs, and student organizations.

Programs

To support the mission of diversity, equity, and inclusion, the office leads many initiatives. They have previously released a strategic plan in fall 2020 [4] that introduced several new programs and goals. Some of these new works include improving scholar and research opportunities, establishing DEI awards, and creating exciting courses on topics like bias, discrimination, anti-racism, and anti-sexism. Some of the interesting and popular undergraduate courses include 85350 Psychology of Prejudice, 82-283 Language Diversity & Cultural Identity, 85-357 Navigating Race and Identity in America: The Role of Psychology in Racial Interactions, and 79-226 African History: Earliest Times to 1780 [1]. These course offerings not only reflect the office's dedication and commitment to DEI but also engage undergraduate students in new and fun ways.

The office has also made notable accomplishments such as expanding the amount of admissions counselors to prioritize diversity, joining organizations like Vibrant Pittsburgh and the LEAP program, and expanding professional development opportunities. Lastly, the Dietrich College Dean's Office of DEI is responsible for collecting data on all of the DEI events, programs, workshops, and more. Statistics help determine the level of outreach and success of the initiative.

Staff

Associate Dean Ayana Ledford is the point of contact for the Dietrich College Dean's Office of DEI. As the strongest advocate for our project, she is a good person to entrust with the project's long-term health. A problem that she faces is that her current system for communicating with her network of volunteer collaborators is too inefficient. She is often chasing old emails to follow-up with them.

When individual departments share their data externally, they often do so through aggregate reports. Consequently, Ayana sometimes cannot access the more interesting, granular data points. She said this sometimes makes it difficult to measure the success of specific projects. Periodically, she must also request demographic information from the undergraduate admissions office which takes up to a month to compile and send a report.

Dietrich DEI engages the community through either the Dietrich DEI Advisory Committee's or the Dietrich DEI Implementation Committee's monthly meetings. During these meetings they discuss the status of old initiatives, the implementation of new ones, and general ways in which to advance DEI's goals for a better campus culture. We have successfully arranged to sit in on a meeting but unfortunately it takes place after the revised proposal deadline. From talking with Ayana we understand that the committee is made up of volunteers who hope to better integrate diversity and inclusivity into campus culture. Multiple volunteers will apparently also repeatedly request Ayana for the same information, causing a redundancy.

Each department is independently responsible for collecting data on DEI initiatives. Doing so is usually only one part of a department administrator's larger data collection job. If you want to change how a department collects data you need to convince the business managers. They will be responsible for accepting and implementing our software, so it is critical that we sell them on it. Organization members volunteer to collect data for DEI and send it by email to Ayana. These volunteers are part-time, and no formal system exists to teach or enforce a uniform set of standards. When considering the volunteer's key role in data collection, we should build systems that allow them to deliver reports quickly, easily, and perhaps with a reminder system.

While Ayana can champion our solution, she lacks the background required to maintain or administrate a complex technical solution. If our solution needs a developer, we should reach out to Dietrich's Assistant Dean of IT and Operations

Vince Sha. Vince is the logical choice because he oversees the same domain that our project encompasses. In addition, he already works to maintain the school's software, such as Dietrich's content management system, so our solution matches the responsibilities of his job. Vince is also our go-to person for understanding the CMU website's Content Management System.

Dietrich employs staff with every level of technological expertise. We must be mindful for example that some staff are nearing retirement, and will likely find it difficult to adopt tech that changes their workflow.

Tech Infrastructure

Throughout our conversations with Ayana, we learned about the DEI office's technological infrastructure. Most of the office's work occurs outside of the technology, as the office prioritizes interpersonal communication and increasing inclusivity on campus. That being said, the network of committees and Dietrich college department heads is linked through email and weekly staff meetings on Friday, which occur in person. Email is the main technological platform for communication and occurs exclusively on the CMU SECURE network and CMU company computers, as the communication exists only through CMU staff members. Meanwhile, Microsoft Excel is the main platform for storing student information and event information/turnout. Ayana briefly mentioned that the Excel spreadsheets contain tables of student information in regards to undergraduate admission demographic data. Tables are one essential part of the DEI office; containing student information or event logistics in a digestible format is necessary to facilitate easier communication with other departments. Similarly, another format of data that is commonly used by Ayana's office is graphic bar charts, line graphs, and pie charts – mostly made through Tableau. These graphics are used to display diversity statistics to the general public through the CMU website.

The current challenges that Ayana's team faces are not due to the type of technology used, but rather the method that it is used. There is no standardized format to input information, and no standardized way to relay it to other departments. Ayana stated that the technology infrastructure is not advanced and disorganized among the DEI department, so the team could also be benefitted by adding easy-to-use software. Specifically, a centralized database, for example, a Ruby on Rails database, where departments could punch in data and send it to Ayana in real-time, would make the network of the DEI department stronger. Since all relevant staff members use CMU-wide OS hardware and software, it would not

be particularly challenging to broadcast such tools, whose accessibility can be accessed through the CMU tech catalog, to the entire office.

Tech Management

The Diversity, Inclusion, and Equity office is a subset of Carnegie Mellon, and so the majority of the technology is managed by Computing Services. The formal process for logging problems is either emailing IT support, calling IT support, or filing a self-service ticket. The CMU technology is mostly managed by this service, and their office fixes issues promptly. Data security threats, breaches, and troubleshooting are exclusively handled through Computing Services. The technology is monitored regularly, updated, and backed up regularly, though department-specific issues need to be explicitly defined in a service ticket. Though the technology management is managed by a different CMU entity, the diversity committee members are responsible for managing their own technological tools to communicate with other DEI departments. Specifically, they are responsible for forwarding event information to Ayana and the Dean's office and use Microsoft Excel at their discretion to format their data. Ayana handles the data from the different committees on a general scale to assess which initiatives have the most turnout and impact on the CMU community.

Tech Planning

In our preliminary meeting with Ayana, we had a little discussion about the technology planning of the DEI committee, especially since the overall communication goals were mentioned more than the technology. Budgeting for the DEI committee is likely done by the Dietrich Budgeting and Planning Committee and Information Technology Committee. Other than that, other CMU technological leaders, such as CTO Stan Waddell, likely collaborate with Dietrich's heads to create the budget. Utilizing technology for long-term diversity initiatives has barely been addressed in the 2-year DEI strategic plan, but it is likely that Ayana plans for the communication and diversity efforts to be supplemented by technological tools. Coordinated technology planning is essential for ensuring that the DEI team can address their general goals and work together to share data. Organizations similar to the Dietrich DEI office, such as other college DEI offices, rely on the same hierarchical structure in CMU to allocate the budget and resources to complete their diversity initiatives.

Communication

Dietrich DEI is a low-tech office. Ayana directly oversees five employees but the office is self-led and relatively autonomous. Before the pandemic, in-person discussion used to be part of communication. All in-person meetings, such as the DEI implementation committee meeting, monthly advisor meeting, and subcommittee meeting, have been moved for the duration of the pandemic to Zoom. They use Google Drive as a shared file system when collaborating. Dietrich DEI emails volunteer organizations to collect data from initiatives they are collaborating on. She has an email list-serve set up by Joanne Ursenback for mass communications. A newsletter is sent out once a month. While the website is updated regularly, Ayana feels that not enough people read the news and roadmaps posted to it.

Information Management

As the Dietrich College Dean's Office of DEI implements strategies and manages initiatives, they work with large amounts of data. For instance, DEI events produce data such as the number of attendees, demographics of attendees, the number of hosts, and more. This information is critical for understanding the outreach levels and assessing the impact of their initiatives. Currently, there is no information management system to house all this critical data. Instead, the associate dean for DEI manages emails and excel spreadsheets from multiple faculty, staff, and volunteers. Thus, there is a lot of additional effort required to gather the information and the overall process is inefficient and time-consuming for all parties involved.

Business Systems

Carnegie Mellon University uses Workday so that employees can log their hours and the HR department can manage payroll. Workday has been relied on as Carnegie Mellon's payroll software since 2015. While not perfect, the website's easy design makes it ideal for novice users. While largely reliable, managers and HR must resolve issues when their employees incorrectly submit hours.

Dietrich DEI uses Workday to have their employees log their hours. Human Resources administrates the system to help process payroll. They can see the hours logged by each employee. Program heads such as Ayana can view the hours logged by employees they directly manage. HR also provides in-person and online assistance to users struggling with Workday. Dietrich volunteers working with the

offices do not log their hours in Workday and are not compensated for their work. In Dietrich, every department runs on its budget. Dietrich DEI's office only requires official budget plans for if they allocate more than 5,000 to 10,000. For those matters, Ayana looks to Richard Scheines for approval.

Changing the accounting software used by Dietrich's DEI office would disrupt Carnegie Mellon's existing payroll system. CMU needs its accounting managed by a centralized system, otherwise tracking accounting fees for every organization becomes impossible. Doing so would create a tech silo and disrupt CMU's accounting, which outweighs the added simplicity and functionality of any software alternatives.

Dietrich's DEI office also spends its money funding initiatives across CMU. They also persuade the relocation of Dietrich funds. For example, they want to repurpose the Dietrich Presidential Fellowships to motivate admitting a more diverse graduate student body.

Community Partner Project Opportunity

Project Opportunity

The biggest challenge the Dietrich DEI committee faces is collecting, storing, and accessing committee data in a centralized format. Currently, when events and initiatives are led by committees, there is no defined way for them to forward attendance and participation to Ayana's office. Ayana often reaches out to the different departments in Dietrich herself to gather the data, which has proven to be extremely inefficient and time-consuming. She also reaches out to the master's admissions office, as demographic data about their admission is much harder to obtain than the undergraduate admissions data. Their primary system consists of Excel spreadsheets sent via email, which can easily get lost and difficult to synthesize with other committees.

Establishing a more efficient and organized system for collecting data on DEI initiatives is important because currently, the Dietrich College Dean's Office of DEI cannot accurately assess impact. They cannot determine how well the programs are going nor what programs are happening within separate departments. They are missing the opportunity to collect crucial data and to measure the success of all their hard work.

A technology solution that allows committees to easily store data, describe their events, and allows Ayana's team to instantly view and organize it would greatly benefit the organization. Additionally, by having a single source for DEI data, tools like Tableau could be utilized to seamlessly create graphics to post to the public. By monitoring trends in the data library, Ayana and the Deitrich College dean would know what works best for implementing DEI efforts in Dietrich College.

Proposed Project Vision

Essentially, our proposed project vision involves improving the speed of data collection and data storing within the DEI department.

The biggest pain point that Ayana said she faced was having to manually reach out to department heads to ask for relevant demographic data. Such data is not standardized, not easy to compile/visualize, and is often lost in the chains of emails she has. To combat this, our proposed solution preliminarily involves building a standardized form for her to send out to department heads via email. Then, if the form data is automatically added to an Excel spreadsheet (stored somewhere online like Google Drive), then the manual process of Ayana forming questions to ask and compile answers is completely eradicated. A labeled, accessible spreadsheet could also make Ayana develop visualizations of demographic and event attendance distributions.

- Value: This solution directly addresses the main pain point that Ayana faces, which is slow communication. Sending a form link, as opposed to taking time to curate questions and then send them over, will reduce the amount of time it takes Ayana to communicate with department heads. Likewise, a pipeline that transfers form data to a spreadsheet will reduce the amount of time it takes for Ayana to compile the data into a place other than her email. A simple Google Drive folder containing the form, responses, and a compiled spreadsheet will allow Ayana to interpret and visualize the data however she wishes to. The main stakeholders for this solution include the department heads, who will benefit from a standardized form with pre-defined data to collect, and Ayana/Dietrich deans, who would now have a centralized area to view data and potentially visualize it.

- Alternate Solution: A possible alternate solution to helping the DEI team with data collection/storage is building a database from scratch, and having department heads enter information directly into it so no emails are sent at all. Though this is probably the most efficient solution for time, it is not feasible for many reasons. Firstly, Ayana emphasized that the department heads are not tech-savvy, and are most comfortable with sending information and communicating via email and Google Suite. Additionally, a custom-built application would require constant maintenance and bug fixes, which would likely be burdensome for Ayana or the CMU IT department.

Expected Outcomes

Figure 1: Actor list

Actor Name	Level	Description
Volunteer Data Collector	Novice	Volunteers from across Dietrich who agree to record data on DEI related initiatives in their department. Data collectors might also be hosting the event they collect data on. Assume a low tech competency, and an inherent unwillingness to change workflows.
Department Administrator	Advanced User	Collects data for a single department, including non-DEI related data points. Depending on the size of the department, admins might also be volunteer data collectors.
Business Manager	Novice user, advanced understanding of product value	Business managers are unlikely to be users of the proposed solution. However, we must sell the value of the solution if we hope to have a department change their workflow to implement it.
Data Analyzer (includes Ayana, Associate Dean for DEI)	Medium User	Data analyzers want to study data related to DEI initiatives. They hope to use the data to improve current DEI initiatives and devise new ones. They want to be able to filter the data around what's relevant to their current inquiry.
Developer (possibly)	Advanced User	Responsible for long-term maintenance of the solution, as well as adding new features.

Figure 2: Main Features & Prioritization

Features	User	Details	Priority
Easily share collected data with the wider DEI ecosystem.	Volunteer Data Collector, Admin	• Must be user-friendly and accessible. • Reduces the amount of follow-up emails. • Possible implementation: data collection form.	**Essential/High**
Data storage solution that stores data in	Data Analyzer, Developer	• Currently, data is silo-ed across multiple departments. • Storage could help facilitate	**Essential/High**

a single, accessible place.		analysis on long-term trends. ● Possible implementation: single database.	
Enforces standards for what data to collect and how.	N/A	● Facilitates comparing data across departments. ● Standards shouldn't limit how departments collect data on themselves beyond DEI. ● Possible implementation: framework for standards baked into the solution.	Nice-to-have
Notification reminders to help sustain conversation between the DEI office and volunteers.	Volunteer Data Collector, Admin, Business Manager, Data Analyzer, Developer	● Reminders that a message hasn't been followed up on, or data is needed that hasn't yet been collected. ● Reminders will lessen the communication gap between the DEI office and other organizations. Reminders can help sustain communication through until the completion of the objective, allowing DEI to move quickly through any multi-step collaboration. ● Example: reminder sent to a volunteer if it's a month after an event and they have forgotten to submit data.	Nice-to-have
Filter / format the database around specific data-points, initiatives, or departments.	Data Analyzer	● A data analyst hopes that the gathered data will help to inform future DEI work & decisions. They must be able to easily analyze all relevant data-points in the relevant context.	**Essential/High**
Easily search database for a specific data-point.	Data Analyzer	● Possible implementation: search bar.	Nice-to-have

Data Visualization	Data Analyzer	• Low priority due to its dependency on data collection and storage systems. • Could integrate existing visualization software like Tableau	Nice-to-have
Modify the data collection standard to collect a new data-point, or stop requesting an old data-point.	Developer	• Data standards should be flexible enough to satisfy future demands, or our solution risks having no long-term future.	**Essential/High**

Figure 3: Major Steps, Expected Outcome, & Baseline Measurements

Activity	**Expected Outcome**	**How Outcome will be Measured**	**Baseline Measurement**
Gathering client context.	List of biggest pain-points, organization details, current workflow, interview all stakeholders.	In the final context meeting with Ayana, we will use our proposal to elicit what we're missing, and have her critique our understanding of the context.	Team shares a solid understanding of what the problem is. Members understand the context well enough to propose an appropriate solution.
Devising the technical solution.	Develop a technical solution that best addresses Ayana's pain-points.	Does the solution relate to the pain-point? If so, how? Why is this solution better at addressing the problem than any alternative?	Team and client agree upon the best technical solution. Should know the solution's overall purpose, the problem it solves, and its features.
Scope features	An MVP with a	Is there enough	A realistic project

down to MVP and work it into the project timeline.	feature list and a time estimate for each one. Time estimate integrated into project timeline to validate feasibility.	buffer time to handle set-backs in our timeline? For example, having us finish implementing features two to three weeks before the final deadline). Estimable tasks.	timeline that attempts to proactively handle possible issues down the road.
Implement new features, test them.	Working, thoroughly tested features that have been seamlessly integrated into the solution.	Test cases; peer review before merging branches to master.	All essential features must be implemented by the end of the project. Hopefully, we will also implement some nice-to-haves that improve the solution's power.
End-user testing & iterating the solution.	Iterate solution based on feedback we get from testing current solutions on end-users.	Interview & come with questions for user; record user while he thinks out loud and uses the software; does new solution address issues raised with old solution?	Invite at least one end-user in for testing for each user type. Ask them clarifying questions and emerge with possible ways to improve the MVP.
Get Dietrich departments to implement the solution.	Convince business managers to adopt software and train volunteers and admins to collect data.	How many departments agree to implement our solution.	Get as many departments as possible to adopt our solution.
Handover solution to client	Clients will receive enough training and/or documentation to sustain and support the solution across the long-term. Handover all	Can the client re-explain our solution back to us so we know they understand its function and value? How well can the client implement	Client should understand enough to independently manage the solution going forward.

	software to the developer.	change (e.g. can client add new data-point without having to go externally for help)?	

Addressing Sustainability Needs

Ayana thinks that the most meaningful DEI initiatives are the ones that encourage a long-term commitment to change. Our solution must be sustainable if it is to be impactful. We plan to begin preparing for hand-over as soon as possible. We have a meeting with Vince Sha where we hope to ask if he will act as a developer after handover. If we use a Github repository we will include a README for setting up the project environment. Our code, if it exists, will be well-commented. In terms of training volunteer data collectors, we hope the UI will be intuitive enough to guide them through the software. However, we can include user documentation if necessary. We also want the client to re-explain the project back to us because once we are gone it will be difficult for us to clarify any confusion.

User Stories

1. As an Associate Dean for DEI, I want to know how many people attended an event so that I can gauge interest and outreach success.
2. As an Associate Dean for DEI, I want to know the demographics of the people who attended so that I can measure the impact of our initiatives.
3. As an Associate Dean for DEI, I want a standardized way of data collection across departments so that I can save time retrieving data.
4. As a Department Head, I want to know the biggest event from another department so that I can implement similar initiatives in my department.
5. As a Data Gatherer, I want to fill out a form so that I can easily contribute data to DEI.

Feasibility

The proposed project is feasible with the allocated time in the semester. After discussing with the client, we have prioritized the data collection and storing processes since that will create the foundation for future data visualization. Data resources are available since Ayana has access to past admissions data and DEI information from varying departments. The project fits the skills of our team since

we have all taken courses to build on our technical and data skills. We all have experience with web development and SQL to implement user-friendly software and process data.

Since Ayana's current responsibilities include retrieving data and gathering reports, she can continue to maintain our proposed solution. Additionally, she is familiar with excel spreadsheets and working with data. For more technical information, Vince and Nicole are more proficient and familiar with the technical details, so they may also maintain and fix any future issues that arise. Lastly, both Vince and Nicole may be able to give us insight into the Content Management System and data they usually manage.

Risk Management

1. **Solution Complexity Risk**
    a. Description: The proposed solution may be too complex for the average user to use and maintain.
    b. Likelihood of Occurrence: Medium; Some staff members are less tech-savvy.
    c. Magnitude of Potential Loss: High; Users may be discouraged and not make use of the solution at all.
    d. Risk Management: Team will develop a solution with user-friendly principles and the capabilities of the staff in mind. Team will also user-test the proposed solution and design iteratively.
2. **Scope Creep Risk**
    a. Description: Scope creep may develop and there may be too many features that the client and team desire.
    b. Likelihood of Occurrence: Low; expectations and known exclusions are clarified between the team and client.
    c. Magnitude of Potential Loss: Medium; Team would lose focus of priorities due to scope creep and the essential features may not be fully developed.
    d. Risk Management: Team will prioritize tasks and further clarify what is in scope and out of the scope with the client. Both sides will review the successes of the week to keep progressing.
3. **Solution Interference Risk**

a. Description: The proposed solution may interfere with the limitations of the Content Management System (CMS) and cannot be implemented.
b. Likelihood of Occurrence: Medium; Client revealed that the CMS is not very flexible and has limitations on what can be done such as uploading certain content.
c. Magnitude of Potential Loss: High; Entire solution may not be usable due to conflicts with the CMS.
d. Risk Management: Team will meet with the technical members from the client-side to better understand the CMS capabilities and limitations. The team will also keep feasibility in mind when developing the solution.

4. **Security Risk**
    a. Description: Team may not have access to real data from the undergraduate or graduate admissions teams due to security guidelines.
    b. Likelihood of Occurrence: Medium; Real data may be sensitive and available to the public eye such as our team.
    c. Magnitude of Potential Loss: Medium; Team will have a harder time understanding the structure of the data and working with data collection, storing, and accessing, but dummy data may be available.
    d. Risk Management: Team will inquire about data accessibility early on and understand what data is available and what is not. The team may also want to create dummy data to analyze and experiment with.

5. **Data Retrieval Risk**
    a. Description: Retrieving data from the undergraduate or graduate admissions team may take months.
    b. Likelihood of Occurrence: Medium; Client revealed that gathering data is very time-consuming and is a long process from the master's admissions side.
    c. Magnitude of Potential Loss: High; Obtaining this data may take longer than the project timeline and may be crucial to understanding the overall data process.
    d. Risk Management: Team and evaluate the significance of this data together and determine how long it would take. The team would also

get an early start and see if this data is feasible to include in the solution.

6. **Sustainability Risk**
    a. Description: Our proposed solution may not be sustainable or useful long-term.
    b. Likelihood of Occurrence: High; Current solution is not sustainable since no one can maintain or keep up the technical solution.
    c. Magnitude of Potential Loss: High; Our solution would be useless if no one could use it long-term.
    d. Risk Management: Team will reevaluate the client problem, project opportunity, and proposed solution to create a sustainable solution for the organization.
7. **Addressing Wrong Problem Risk**
    a. Description: Our proposed solution may not be addressing the real help that the client needs from us.
    b. Likelihood of Occurrence: Medium; Current solution addresses the data collection and data storing issues but the client may have a larger or deeper issue that would be more beneficial if solved by the team.
    c. Magnitude of Potential Loss: Medium; Our solution would be helpful but not really target the largest pain point by the client.
    d. Risk Management: Team will reevaluate the client's problem, project opportunity, and proposed solution to really understand the root of the problem and the problem space so that we are helping the most important and provide the most benefit to the client.

Resources

1. https://www.cmu.edu/dietrich/about/dei/index.html
2. https://www.cmu.edu/dietrich/about/dei/committees.html
3. https://www.cmu.edu/dietrich/about/dei/strategic-plan.html
4. https://www.cmu.edu/diehttps://www.cmu.edu/dietrich/about/dei/strategic-plan.htmltrich/about/dei/docs/diversity-equity-inclusion-strategic-plan.pdf